

Cryptography

Discrete Mathematics
Courtesy of Andrei Bulatov

Previous lecture

- *Modular arithmetic and inverses*

- *Applications*

- *Systems of linear congruences*

- *The Chinese Remainder Theorem*

- *The system of linear congruences*

$$x \equiv a_i \pmod{m_i}$$

where each m_i, m_j are relatively prime has a unique solution

$$x \equiv \underline{a_1 M_1 M_1^{-1}} + \dots + a_k M_k M_k^{-1} \pmod{m}$$

$$n = 105$$

Sun Tzu's Puzzle

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$M_1 = 5 \cdot 7 = 35$$

$$M_1^{-1} = 2$$

$$M_2 = 3 \cdot 7 = 21, M_2^{-1} = 1$$

$$M_3 = 3 \cdot 5 = 15, M_3^{-1} = 1$$

$$X = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1$$

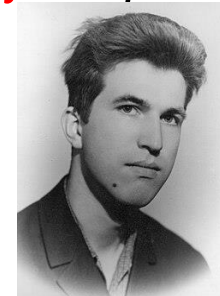
$$\equiv 140 + 63 + 30$$

$$\equiv 233 \pmod{105}$$

Diophantine equations

- *Systems of linear congruences are examples of a larger class of equations of massive importance in mathematics: **Diophantine equations***
- *Diophantine equations are equations of the form*
$$ax^n + by^m + cz^p + \dots = d$$
where $x, y, z, \dots$ are unknown integers to be solved for
- *Linear congruences are Diophantine equations of the form*
$$\underline{x = y + mz}$$
- *Hilbert's tenth problem*

*Devise an algorithm which can solve **any** Diophantine equation*
- *Hilbert's tenth problem was closed in 1970 by Yuri Matiyasevich, who showed **no such algorithm can exist***



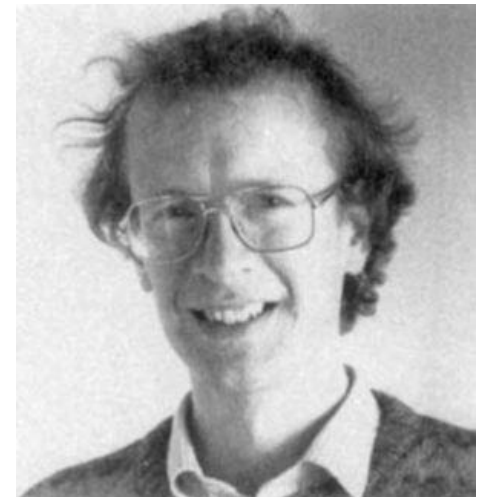
Fermat's Theorem

- **Fermat's Great (Last) Theorem.**

*For any $n > 2$, the equation $x^n + y^n = z^n$ does **not** have integer solutions $x, y, z > 0$*

- *It had remained unproven for 358 years (posed in 1637, proved in 1995)*

- *Andrew Wiles proved it in 1995 with a **100 page proof** using group theory, Galois theory, and tools from algebraic geometry*



Fermat's Little Theorem

● Fermat's Little Theorem.

If p is prime and a is an integer not divisible by p , then

$$\underline{a^{p-1}} \equiv 1 \pmod{p}$$

● *Clearly, it suffices to consider only residues modulo p .*

$\mathbb{Z}_5$

.	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Fermat's Little Theorem (cntd)

- Recall the definition of Euler's totient function:

$\varphi(m)$ = # of integers up to m which are relatively prime to m

- Euler's theorem (Improves Fermat's little theorem)

For any relatively prime integers m and a ,

$$a^{\underline{\varphi(m)}} \equiv 1 \pmod{\underline{m}}$$

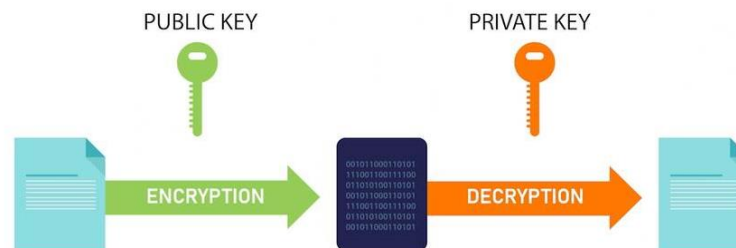
- Example: $\mathbb{Z}_8$

$$3^4 \equiv 1 \pmod{8}$$

$$5^4 \equiv 1 \pmod{8}$$

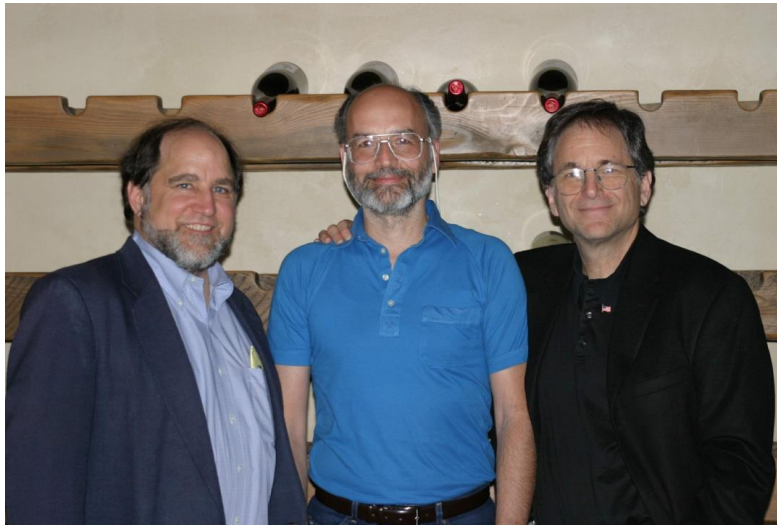
Public Key Cryptography

- The internet is built on **cryptography** --- private data (like banking) is **encrypted** so that an eavesdropper can't read it
- The Caesar cipher from last class is an example of encryption
- In it, a **key** is used to both encrypt and decrypt
- If **the same key is used to encrypt and decrypt**, anyone who can send you messages can also **read** your messages!
- **Public key cryptosystems** solve this problem by using different keys for encryption and decryption:
 - a public encryption key, and a private decryption key



RSA Cryptosystem

- *The most famous and widespread pubkey system is RSA*
- *RSA stands for the names of the inventors: Rivest, Shamir, Adleman*



*From left to right:
Ron Rivest
Adi Shamir
Len Adleman*

- *RSA is defined by*
 - *a number $n = pq$, where p and q are large primes (e.g. 512 digits), and*
 - *a number e relatively prime with $\varphi(n) = (p - 1)(q - 1)$*
- *n and e are the public encryption keys, while p and q are private*

RSA Encryption

● Let M be the plaintext message, written as a number

● The encryption ciphertext is the residue

$$C \equiv M^e \pmod{n}$$

● Example.

- Let $p = 43$, $q = 59$, so $n = 43 \cdot 59 = 2537$
- Choose $e = 13$ which is relative prime with $\varphi(n) = 42 \cdot 58 = 2436$
- Encrypt the message STOP with the public key n, e

● Solution.

Step 1: Translate the letters of STOP into numbers in groups of 2: 1819 1415

Step 2: Encrypt each group of 4 by: $C \equiv M^{13} \pmod{2537}$

$$1819^{13} \equiv \underline{2081} \pmod{2537}$$

$$1415^{13} \equiv 2182 \pmod{2537}$$

Step 3: Send the ciphertext 2081 2182

RSA Decryption

● Let d be the inverse of e modulo $\varphi(n)$

– Note: d exists because e is relatively prime with $\varphi(n) = (p-1)(q-1)$

● Claim: if $C \equiv M^e \pmod{n}$, then $C^d \equiv M \pmod{n}$

● Proof:

Since $de \equiv 1 \pmod{\varphi(n)}$, $de = 1 + k \varphi(n)$

Now $C^d \equiv (M^e)^d \equiv M^{de} \equiv M^{1 + k \varphi(n)} \pmod{n}$

By Fermat's little theorem,

$$M^{k \varphi(n)} = (M^{\varphi(n)})^k \equiv 1^k \equiv 1 \pmod{n}$$

So $C^d \equiv M \cdot M^{k \varphi(n)} \equiv M \pmod{n}$

Example

- Let $p = 43$, $q = 59$, $e = 13$ as before
- We want to decrypt the message 0981 0461
- Solution:

Step 1: compute $d = 937$ as the inverse of 13 modulo 2436

Step 2: decrypt each block as $M \equiv C^{937} \pmod{2537}$

$$0981^{937} \equiv 0704 \pmod{2537}$$

$$0461^{937} \equiv 1115 \pmod{2537}$$

Step 3: Convert back to letters:

07 04 11 15

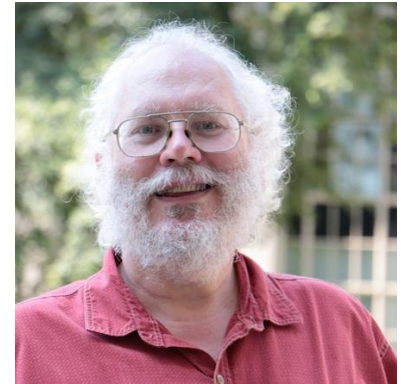
H E L P

Why RSA Works

- RSA is predicated on *factoring large numbers being HARD*
- Specifically, we can decrypt RSA knowing just the inverse of e modulo $\varphi(n)$, but computing $\varphi(n)$ requires *knowing the prime factorization of n*
- Best current (classical) algorithm for computing factorizations:
Number field sieve, running time $O(\exp(\sqrt[3]{m}))$
where m is the length of n
- For $m=400$, this is *billions of years*
- RSA Challenge: Factor a published number $n=pq$, get a prize
- The current record is RSA-250. Next one is RSA-260:
221128255295296664352810852550262309276120895024700153944137483191288229
414020019865127297265697465990859003300314000511707422045608592763579537
571859542988389587092292384910067030341246205457845664136645406842143612
93017694020846391065875914794251435144458199

Quantum computing

- With *quantum computers*, the situation changes drastically
- In 1994, Peter Shor proved that a computer *operating on the principles of quantum (rather than classical) mechanics* can efficiently (i.e. quickly) factor larger integers
- The quantum computers in existence today can NOT factor RSA-2048 yet...
 - ... But maybe in 10 years time?
- Many current public key cryptosystems are also now broken (in theory) by quantum computers
- *Post-quantum cryptography* develops quantum-safe cryptography so that secrets today will not be public tomorrow!



Practice

Exercises from the Book:

7th edition: 17, 20, 31, 33, 35 (page 284 – 285)

8th edition: 17, 20, 31, 33, 35 (page 301)